

Overseeing Many LLMs: A User-Controlled Review Queue for Concurrent Agent Workflows

Anonymous Submission

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Abstract: Concurrent language model workers produce outputs that a user must inspect, defer or release while other work continues. We argue that these outputs should be individually reviewable, versioned requests with stable evidence and visible unfinished work. Our implemented desk connects asynchronous workers to a common review-and-release protocol, with a central queue and optional source sessions. A completed software evaluation uses 24 TAT-QA financial-report contexts, 145 original questions and 290 actual model answers. All 13,236 scenario events replay consistently, and active-review checks preserve the inspected output during arrivals. Ideal scripted corrections yield a mechanical completion tie, not a human benefit. Initial exact match is 61/290. A subsequent audit restores seven parser-lost displays, reducing empty or unsupported answer fields from 38 to 31 without changing exact match. The runnable formative pilot has two nine-minute queue/session blocks and a separate four-minute source-only diagnostic. No participant observations exist. This position paper contributes a specified oversight workflow, its real-data software characterization and an executable evaluation protocol. Whether source sessions improve correct review or experienced control remains to be measured.

1 INTRODUCTION

A person supervising several language model workers needs to know which outputs require a decision, what evidence supports them, and what remains unfinished. An arriving answer should not replace the source they are reading. A postponed question should retain its draft and history. If an agent revises an answer after approval, the new artifact should not silently inherit permission to be released. These are concrete requirements for a workflow in which the user retains authority over release.¹

We investigate how a person can organize, inspect and resolve competing actionable requests while retaining visibility of unfinished work and control of release. Our application is financial question answering. Several independently prompted task workers use one shared model to answer original questions about real financial-report tables. The user can inspect the source, approve one output, supply a correction, reject it or defer it. A shared source supports navigation, but never implies that two questions share a correct answer.

The design favors a strong simple queue over elaborate allocation or automatic correction propagation. Earlier internal studies found that more complicated

ordering and uncertain correction reuse did not consistently improve outcomes. Those findings motivate separate decisions, direct evidence and explicit user control. They do not validate this interface.

Our position is that outputs awaiting action should be explicit review objects, rather than only entries in an agent transcript. We operationalize that position in a review-and-release protocol, a desk with live and saved-output adapters, and a frozen real-data software evaluation. The protocol makes accounting, focus stability and version authority testable under concurrent arrivals. A completed preparation package makes a formative comparison of the central queue and optional source sessions runnable, with fundamental approval semantics common. We present this as a position supported by an implemented system and measured software behavior. Human review quality and experienced workload remain open empirical questions.

2 RELATED WORK

Agent dashboards and intervention interfaces already have close precedents. AgentGUI supports concurrent desks, activity views, intervention and saved sessions. Its eight-participant evaluation concerns trace

¹AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

comprehension, while a separate experiment tests automated steering (Zhao et al., 2026). That evidence is stronger than ours on human interaction. Our question concerns which individual output may be released after inspection; we have not compared the two systems.²

AGDebugger provides queues, pause/step controls, message edits and checkpointed conversation forks. Its studies examine error identification and steering of multi-agent teams (Epperson et al., 2025). These mechanisms establish that queues, history and intervention are not new contributions by themselves. Our narrower focus is the lifecycle of individual outputs awaiting a user’s release decision.

Human multi-robot supervision also has a substantial history of attention allocation. Chien et al. compare open-queue and shortest-job-first presentation (Chien et al., 2018). Research on interruption recovery motivates retaining a postponed task’s context (Iqbal and Horvitz, 2007). We apply these established principles without interpreting queue length, token counts or scripted action time as mental demand. These predecessors do not establish that our workflow bundle is better. Nor does our review establish that they lack every analogous safeguard. The narrower research position is to make release authority, active evidence and unfinished work explicit, then evaluate them with the same source tasks and decision rules. B is a competent ordinary queue, not a deliberately fragile baseline. A/B/C are configurations of our application, not reproductions of the cited systems.

3 REVIEW PROTOCOL

3.1 Requests, Sources and Versions

An offered task records a worker identifier, original question, source identity and hash, provenance, and any explicit dependencies. An actionable request is created when its proposed answer arrives. It records the artifact version, receipt time, requested decision and review status. Generated explanations and citations are labeled as generated. The original source table and paragraphs remain available even when an answer or its citation format is incomplete.³

The task lifecycle is offered, admitted, started and received. A received request is queued, active, deferred, resolved or withdrawn. Artifacts are immutable version records. A revision supersedes the

²AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

³AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

current version without deleting earlier artifacts, decisions or release history. The application does not infer dependencies from similar wording. In the TAT-QA application, source identity is used only to organize related questions.

No financial deadlines are invented. The protocol accepts a deadline or priority only with an explicit origin, such as a user’s instruction. Without such metadata, the next-request recommendation is stable FIFO. Explicit deadlines use EDF within priority. Once a review begins, the active artifact is pinned until the user acts. The experimental observation cut-off is not a business processing deadline.

3.2 Actions and Authority

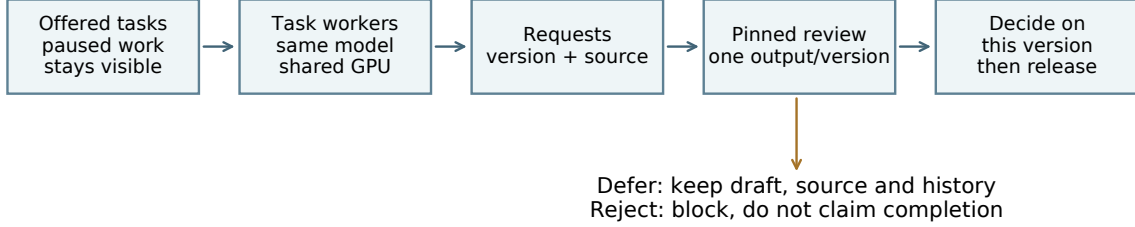
Approval authorizes one exact request/version. Release is a separate action and requires approval of the current version. A user correction creates and approves a new version of that answer only. Separating approval from release lets a user record an inspection decision while withholding the artifact from the local released set, for example while checking another question. Release here is an application record, not an external financial transaction. The extra action may itself cost effort; its usefulness has not been measured. Rejection blocks the answer and resolves the review, but does not complete the underlying question correctly. Deferral retains the source, draft, note and history, with an optional return time. Withdrawal is explicit and does not erase prior events or undo any previously recorded release.

When a revised artifact arrives during review, the old source and answer remain pinned. A visible warning offers an explicit refresh. An attempt to approve the older version is refused. After refresh, the new version requires its own decision. A group contains at most three already received questions with the same source identifier. Grouping never combines their answer fields or approval authority.

Admission control pauses new task starts. It does not interrupt an in-flight model request or prevent that request’s result from arriving. All offered tasks remain in the outstanding-work account, including those that have not started. This distinction prevents an apparently smaller queue from concealing delayed work.

3.3 Atomic Events and Limited Properties

Commands are processed under one lock. Each includes an event ID, action and payload. A repeated ID with identical content returns the earlier receipt.



New arrivals update pending work. A revision never inherits an approval.

Figure 1: Specified software workflow. Every output has a separate version and release decision. Source sessions change navigation, not answer authority. This diagram describes the protocol rather than a measured human effect.

Reusing an ID for different content is refused. Before applying a transition, the controller checks its prerequisites. A refused transition restores the prior logical state. Each command is recorded with a sequence, timestamps, result and state hash.

Let O be offered tasks, U unstarted tasks, F in-flight tasks and R received requests. The implementation checks

$$O = U + F + R. \quad (1)$$

Received requests are similarly partitioned into queued, active, deferred, resolved and withdrawn states. No accepted transition deletes a request. By induction over transitions, every offered task therefore remains accounted for. Immutable version records and copied active snapshots imply that a new arrival cannot change the active reviewed artifact. A release check requires the current version and its own approving decision, so a revision cannot inherit approval. Grouping changes only membership and cannot authorize a sibling output.

These are structural properties under the implemented protocol. They do not imply answer correctness, user comprehension or low workload. All releases must pass through this controller. The local single-user server is not a distributed consensus service or an authenticated public deployment. Journals support recovery and replay, but are not claimed as an fsync-backed crash-atomic database. Recorded logical events replay deterministically; fresh model outputs and timing measurements need not.

4 SYSTEM AND INTERFACE CONDITIONS

The backend has asynchronous task-worker and replay adapters using the same commands. Three independently prompted workers share the existing GPU service. The server processes model requests serially under its configured capacity, while worker waiting, request delivery and user interaction proceed asynchronously. An agent here is a task worker, not a separate model architecture.⁴

The browser polls public state and updates pending work without replacing the active answer, source or text controls. It refuses older state sequences and protects against delayed responses from a prior session. User actions and displayed request/version identities are logged. A deferred review restores its note and source. Saved sessions can be restored without launching workers. The ordinary interface never loads reference annotations. A stable version prevents a decision from being attached to content that changed while it was being inspected; it does not establish that the inspection was adequate.

Condition A presents worker conversations with unread indicators, navigable threads and source access. B presents a central conventional queue with one stable active review. C adds optional bounded source sessions and admission controls. Pending work remains visible in all three. All conditions share source access, exact-version approval, explicit release and deferral semantics. Release guards are identical across conditions.

⁴AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

The prospective primary contrast is B versus C. A is a contextual baseline. C is a workflow bundle rather than an isolated batching mechanism. The software comparison measures its behavior under declared scripts; it does not estimate how people choose groups, defer work or use the pause control.

5 REAL-DATA EVALUATION DESIGN

5.1 Sources and Generation

TAT-QA provides real financial-report tables and text with human-written questions and reference annotations (Zhu et al., 2021). Worker assignments, source bundles, arrival schedules and reviewer scripts are constructed for this study. This is not a field study of financial analysts.⁵

We froze the first 24 eligible unused public test-gold contexts in source order. The audit excluded all 36 contexts used in prior development or evaluation, matching normalized tables and long paragraphs, and detected cross-split source duplicates. Selected contexts have four to eight questions and a source rendering within the existing context limit. All 145 original questions are retained. There are no reliable complete report identifiers, so source contexts cannot be assumed to represent independent reports or companies. Public benchmark exposure to the model is not ruled out.

Two replicas produce 290 initial answers. The short interface requests an answer, scale and optional explanation or citations; it does not require an executable computation schema. The model may select a bounded arithmetic expression for a calculator to evaluate. No reference operands or answers are inserted. Four predetermined source rechecks produce actual revision outputs. Prompts, source selection, generation settings, failure handling and scenarios were committed before fresh generation.

The service is pinned Qwen2.5-7B-Instruct in BF16 on one RTX 6000 Ada, with zero CPU offloading. Temperature is 0.3, top- p is 1, and output allowance is 230 tokens within a 2,048-token model context. Seeds are recorded. The saved model and source provenance, request accounting and reproduction commands remain in the evidence package. No new inference was used to prepare the formative pilot or this handoff. None of the original answers was replaced because of its score.

⁵AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

5.2 Matched Software Scenarios

Twelve constructed bundles each contain two source contexts. Both replicas run in A, B and C at lower and higher arrival concentration, giving 144 scenarios. Every task is offered at time zero. Starts become eligible at intervals of 0.15 or 0.015 seconds, followed by a 0.04-second delivery delay. At most three replay jobs are in flight. These are compressed wall-clock test schedules, separate from observed GPU latency. They are not assumptions about human reading speed.

A script selects one request, waits 0.095 seconds and performs a question-specific ideal inspection. It approves a correct answer or creates an annotation-based correction, then explicitly releases that version. The first request is deferred once for 0.35 seconds. C pauses new starts from 0.25 to 0.65 seconds and may retain up to three source-related questions. Actual saved rechecks enter the declared replica-zero scenarios at 0.8 seconds. Every scenario ends at 2.2 seconds.

Methods share saved answers, sources and eligible-start schedules. C’s pause can change arrival times, but never the offered denominator. All attempts and failures are retained. Bundle-level paired summaries average generation replicas across 12 bundles; they characterize the scripts, not people. A separate live demonstration holds one review while real answers arrive and tests a version change without showing annotations.

6 RESULTS

6.1 Answer Quality and Scripted Completion

The initial model achieves 61/290 exact matches (21.0%), a mean token F1 of 22.61%, and 59/290 joint answer/scale matches. The historical projection has 38 empty or unsupported answer fields. All returned bodies are parseable JSON. Optional citations supplied in a string instead of a list trigger 157 legacy formatting flags without removing the readable answer or source. This distinction matters when interpreting representation validity.⁶

All conditions release all 290 answers per load after ideal scripted correction (Table 1). The script’s capacity is ample at the final cutoff. B and C therefore tie on completion in every paired bundle, with a degenerate script interval of zero. This is a mechanical result, not human equivalence. The low initial

⁶AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

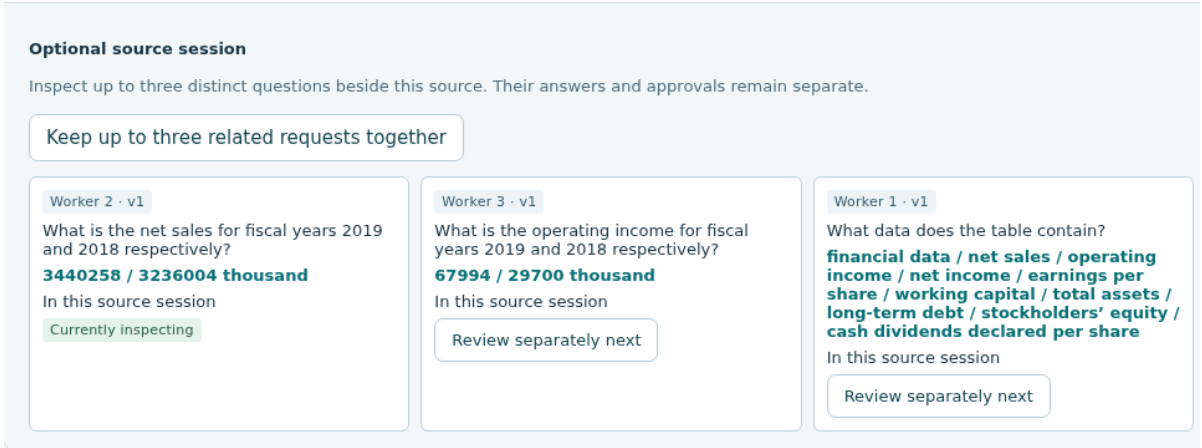


Figure 2: Actual interface capture during scripted replay. Three original questions share one financial source, while their generated answers and review decisions remain separate. The first selected source and first three questions determine this example. The long descriptive answer is retained despite imperfect benchmark agreement. This is software interaction, not a participant observation.

Table 1: Measured scripted outcomes on 24 source contexts and two replicas. Peak is the mean sampled queue maximum. Wait is arrival-to-selection time, including resumed reviews. All releases use ideal simulated supervision.

View	Load	Released	Peak	Wait (ms)
A	lower	290/290	1.04	57.2
A	higher	290/290	10.08	573.6
B	lower	290/290	1.00	57.4
B	higher	290/290	10.08	572.8
C	lower	290/290	2.08	116.4
C	higher	290/290	10.08	583.5

model accuracy makes the distinction between generated work and ideal correction particularly important.

C does not improve scripted throughput. Its lower-load pause increases queue size and arrival-to-selection time. Figure 3 retains tasks that have not started, so pausing cannot improve a completion denominator by hiding work.

6.2 Concurrency, Versions and Latency

All 13,236 scenario events replay to their recorded logical states. The 144 cutoff snapshots and released-answer scores are reconstructed independently. There are no missing, duplicated or misassociated received request IDs in the completed matrix and no unexpected command failures. All 41,722 repeated active-snapshot checks preserve the reviewed output. These are software checks, not independent sources or participant observations.

The live demonstration uses twelve of the initial answers from two source contexts. A six-second scripted active review overlaps observed arrivals while new task starts are paused for two seconds. In-flight generation continues. A fresh model revision creates a new version, and the earlier approval attempt is refused. The demonstration retains all twelve unresolved requests rather than claiming a person completed them. The saved live timeline is available in the evidence package.

Across local measurements, median atomic handler time is 0.74 ms and its 95th percentile is 1.25 ms. Median browser HTTP round trip is 5.75 ms, with a 95th percentile of 15.64 ms. State receipt to JavaScript DOM update takes a median 1.60 ms and a 95th percentile of 6.12 ms. The latter excludes compositor paint. Delivery lateness relative to replay timers has a 95th percentile of 2.17 ms. These boundaries exclude human response and GPU generation time.

Chromium tests pass 62 checks across all interfaces, including typed notes and focus during arrivals, source grouping, deferral, resumption, stale-state rejection, explicit release, connection recovery and a timed study cutoff. Figure 2 shows the source-session component. Earlier harness and implementation defects were repaired with their failed runs retained. The evidence package records those repairs, exact timing boundaries and focused tests. No historical answer or result table was overwritten.

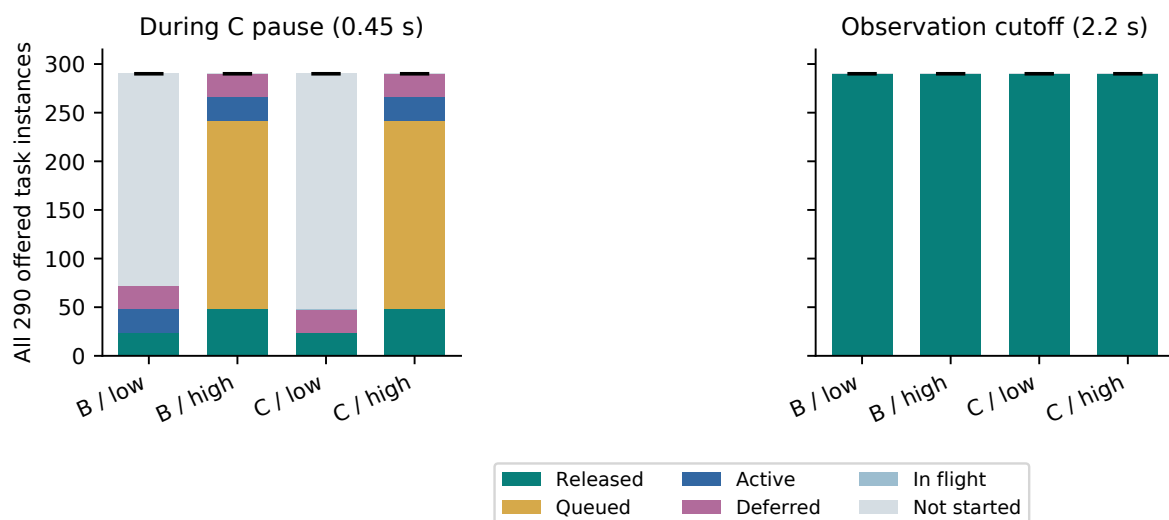


Figure 3: Recorded task accounting during the declared pause and at the final cutoff. The intermediate checkpoint is descriptive. Real model answers and source questions are replayed with constructed timing and ideal scripted corrections. Pausing shifts tasks into the unstated category without removing them.

6.3 Subsequent Answer and Display Audit

Pilot preparation audited all 38 historical empty fields against saved raw responses. Seven nested-array answers had been lost by the display parser. A uniform pilot-only projection restores their existing components, leaving 31 empty or unsupported fields. Exact match remains 61/290 and joint answer/scale correctness remains 59/290. The original 38-empty table and raw answers are preserved. Display recovery is not an accuracy improvement.

Of 24 explicitly empty answer arrays, eight have answer-bearing text elsewhere and four contain partial or qualified text. The pilot exposes the original response in an expandable view without promoting explanations into scored answers. The 157 optional evidence-format flags include 32 jointly correct answers. A format flag is therefore not a correctness label.

A reproducibly stratified audit of 22 nonempty answers includes ten correct controls and twelve scored errors across answer types and source contexts. Nine errors show clear content or unit problems. Three raise unresolved wording, scale or rounding questions. These developer assessments are neither independent adjudication nor an estimate of error causes across all 229 nonmatches. The publicly selected pilot packets retain 36 questions from six contexts, with seven correct replica-zero drafts and five empty

fields. Selection uses public size and question features, not model scores. Review may require reconstructing many answers. The source-only diagnostic is necessary because the usefulness of these drafts is still unknown.

7 RUNNABLE FORMATIVE PILOT

7.1 Design and Treatment

The current prepared protocol is `formative-v1`, not a completed human study. It has separate-source training, two nine-minute B/C blocks, and a four-minute source-only diagnostic. The earlier interface-by-load effectiveness proposal remains a later design option, not part of this pilot. There is one constructed arrival schedule in B and C. The software wrapper, packets, forms, export and analysis have been exercised with explicitly separated software fixtures. There are no participant observations.⁷

Three packets each contain two sources and twelve original questions. Public source length, table size and calculation wording guide selection and matching. The six contexts are reused benchmark material, not fresh independent financial reports. Training uses another source. Each participant sees disjoint

⁷AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

sources in B, C and the diagnostic. Eighteen assignments cross all six B/C/source-only orders with three packet rotations. Coarse matching and counterbalancing do not guarantee equal difficulty or eliminate general learning.

B and C share saved drafts, complete source access, a calculator, stable active review, notes, individual decisions and exact-version release. C additionally provides up to three source-session cards, session-aware next selection and pause/resume of new task starts. Pending work remains visible in both. This is a workflow-bundle comparison, not isolation of a grouping button. All twelve questions are offered at block start, with starts eligible in two six-question waves at 0–20 and 180–200 seconds. Pausing does not remove work from the denominator.

The source-only block offers six original questions from its disjoint packet without model proposals. Its position rotates. It diagnoses source comprehension, calculation difficulty and possible reconstruction burden. Different question and time allocations mean it is not a controlled assistance-effect estimate. A better source-only performance cannot by itself establish that the drafts harm review, nor can a worse performance establish that they help.

7.2 Outcomes and Collection Boundary

The primary descriptions are correct current-version releases, incorrect releases and unfinished work among all offered questions. A correct release requires official answer exact match and scale agreement. Approval without release is separate; rejection is blocked work, not correct completion. Official exact match and token F1 are retained, with scale agreement also reported. A blinded adjudication workflow can add reasoned judgments on released answers, including exact-match controls, without overwriting official scores. No such independent judgments exist.

Interaction records retain deferral, resumption, source revisits, grouping, admission controls and focus interruptions. Select-to-action elapsed time is not continuous reading time. Raw NASA-TLX items measure reported workload (Hart and Staveland, 1988); custom control items remain separate. No ratings are filled by scripts. The participant is the paired B/C unit. Individual trajectories, missing blocks and packet/order imbalance precede any interval. A small formative sample cannot establish superiority, equivalence or population generalization across reports.

The first authorized collection should identify source or unit misunderstandings, incorrect releases, missing-draft burden and failed resumption. A later

effectiveness study needs a meaningful target difference and variance estimates that account for participants and sources. The prepared materials supply neither institutional authorization nor consent. Investigator practice is development evidence, and software fixtures cannot silently enter participant analysis. Any change after practice or formative observations must receive a new development version before subsequent comparison.

8 DISCUSSION AND CONCLUSION

The workflow makes concurrent outputs addressable objects with explicit review and release authority. Its contribution is operational: a user can keep an answer in focus, inspect its original evidence, postpone it without losing context and see what remains before choosing the next task. The implementation and real-data traces make these behaviors inspectable and reproducible.⁸

Real source data matter because the desk must display authentic tables, units, prose and imperfect answers, rather than idealized cards. They reveal missing outputs, substantial calculation errors and annotation ambiguities that a clean fixture would conceal. They do not supply real agent arrival patterns or measured financial-review behavior. The ideal script reveals whether the protocol accepts and records a correction, not whether a person can identify or produce it.

Ordinary queues and version checks remain strong explanations of the system's reliability. No session advantage has been demonstrated. Source sessions may save revisits, or distract from unrelated questions; separate release may support intentional withholding, or add needless actions. Low-quality drafts may make answering directly from the source preferable. The prepared pilot can identify these feasibility issues, but its diagnostic block cannot quantify an assistance effect under matched allocations.

The position is therefore specific and testable: treat concurrent agent outputs as individual, versioned review requests, keep their evidence stable and account for all unfinished work. We provide the implementation and measured software behavior needed to test that position. The next milestone is real interaction evidence from the prepared, authorized formative comparison, before a claim of improved review quality, efficiency, perceived control or workload is made.

⁸AI drafting assistance: OpenAI Codex (OpenAI, 2026). See the disclosure.

AI ASSISTANCE AND REPRODUCIBILITY

OpenAI Codex assisted with implementation, analysis, testing, figures and manuscript drafting (OpenAI, 2026). It is not an author or a participant. Human authors remain responsible for verification and submission. The separate research package contains frozen sources and settings, raw generations, private evaluator projections, event traces, prospective study materials and one-command saved-output reproduction. Historical evidence remains available separately. The review PDF omits author identities and repository links; a public development history may still make the work identifiable. Authors must resolve disclosure and artifact-sharing choices before submission. No participant study has been conducted.

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